



# Proposed CEA Link — December 2026

The existing 20-line link, widened to 24 people — the same turns, the same shape of week, one more cover cycle

Prepared 8 September 2026 · re-rendered 25 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules (Marylbone Roster v24.23) · figures on this page are computed, not typed

## PROPOSAL

# Same Turns

Today's link, widened to 24

CODE **ST-24-B7**

FINGERPRINT **d15e1b74**

BUILT FROM duty table B · seed 7 · 24 lines · 4 cover weeks

✓ **24** lines designed

✓ **35h 00m** a week — exactly the contract

✓ **0** hard-limit breaches

✓ **0** rests under 12h

⚠ **1** fatigue factor present (today: 5)

Five questions a manager asks first, in order. Every figure is computed from the cells; **page 2 says how to read each one**, with today's link and the rest of the folder marked on a scale.

### CAN IT BE RUN?

## Yes

#### the hard limits

0 rests under 12h · longest run 6 of 13 · contract exact

### DOES IT MEET THE DECEMBER SHAPE?

## 4 of 4

#### headcount rules met

four to open, three to close (four on Saturday), five at 22:00, 14 and 10 at the weekend · today meets 0 of 4

### HOW TIRING IS IT?

## 1

#### fatigue factors present, of 25

advisory — present means worth a look, not a breach · today 5 · fewest in the folder 0

### DOES IT FOLLOW THE TRAINS?

## 46.3

#### weekday demand fit — lower is closer

Saturday 32.7 · Sunday 80.6 · today 51.1 · best in the folder 30.7

### IS IT FAMILIAR?

## 15 of 15

#### shift times people work today

nothing new to learn · 18 of 20 weeks are one turn (today 7 of 16)

## 6 in 24

#### full weekends off

25% of the rotation, against 20% today (4 in 20) · most in the folder 25%

## 4 cover weeks

#### lines 1, 7, 13, 19

evenly spaced six lines apart — today's sit at 1, 7, 12, 17

## 2h 33m a week

#### typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

## What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 rather than 20. The brief was to keep the **feel** of the link people work today — a week is one turn, rest days come in pairs, four whole cover weeks spread round the rotation, and the same 06:20 / 08:00 / 14:00 / 15:15 clock times — while meeting the staffing shape agreed for December 2026 (four to open every day, three through to the close, four on a Saturday, five still on at 22:00, fourteen on a Saturday, ten on a Sunday) and every rule the link is assessed against: Chiltern's 13-day limit, twelve hours between duties, the exact 35-hour contracted week, and the ORR's good-practice fatigue factors.

It was not drawn by hand and it was not produced by the workspace's generator, which slides a person's start time through the week. Instead the December 2026 duty table was written in today's shift times, and a search then arranged those duties across 24 lines — over a million candidate arrangements — with the workspace's own hard-limit, rest, contract, fatigue and line-order modules scoring every one, plus a measure of how closely each resembled today's roster. Page 9 says how, and how to reproduce it.

**What it is not.** Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December 2026 headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 7 and 8 say so where it applies. It clears every hard limit the tool can check — twelve hours between duties, the 13-day limit and the exact contract — which is the floor for being worth discussing.

## HOW TO READ THE NUMBERS

# What each figure means, and what good looks like

Every figure on this sheet is computed from the cells. On each scale: this sheet in navy, today's 20-line link as a ring, the best and worst of the other 21 sheets in the folder as gold and grey ticks.

**Three things to hold.** Lower is better on the fit and the fatigue count; higher is better on rest, weekends, one-turn weeks and familiarity. "Present" is not "failed" — the ORR list reports what is there. Today's link is the yardstick, not the target: it meets none of the December 2026 headcounts.

### DOES IT FOLLOW THE TRAINS?

#### Demand fit — weekday

How closely the people on duty follow the trains through an average weekday. For each of its 18 hours, the hour's share of the staff-minutes is set against its share of the traffic; the gaps, in percentage points, are squared and added up — 0 would be a perfect match. The typical hourly gap is the square root of the score ÷ 18: about 1.6 points here, 1.7 today.

**What it does not tell you.** It compares shares, not headcount — adding people evenly to every hour changes nothing — and the station rosters posts that need someone whether the hour carries 23 trains or 5, so a flatter day can be right and still score worse.



Saturday **32.7** (today 45.7, folder best 8.1) · Sunday **80.6** (today 65.9, best 20.9). Each day type has its own curve and window, so a Saturday is only compared with a Saturday.

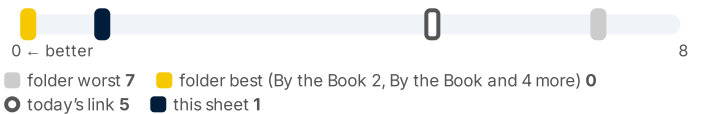
On page 6.

### HOW TIRING IS IT?

#### Fatigue factors present

How many of the 25 roster patterns on the Office of Rail and Road's good-practice list appear anywhere in the rotation — long runs of earlys, short gaps between duties, start times that jump — counted by the workspace's own module.

**What it does not tell you.** A factor present is worth a look, never a breach: the ORR says the list is guidance, not limits. Zero does not mean approved and one does not mean unsafe; page 8 names which, and how big.



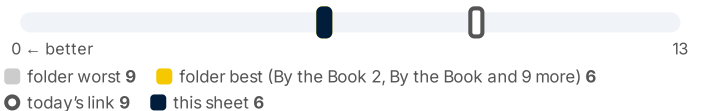
On page 8.

### CAN IT BE RUN?

#### Longest run of worked days

The most consecutive days anyone works in the worst case — a cover week's four duties placed as badly as they can be. Chiltern's limit is 13; the workspace's own target is 6.

**What it does not tell you.** The roster clerk decides where cover-week duties fall, so the as-rostered figure can be lower; page 8 shows both readings.



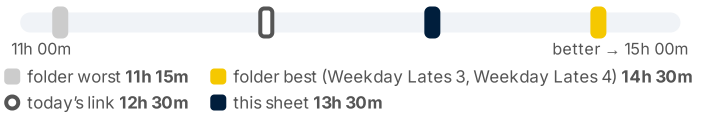
On page 7.

### CAN IT BE RUN?

#### Tightest rest between duties

The shortest gap anywhere between one duty ending and the next beginning — Saturday into Sunday and the last line into the first included. Twelve hours is the floor; a design under it cannot be run.

**What it does not tell you.** It is one gap. A rotation with many 12h05 rests and one with none read the same here; the count under 12h is in the chips on page 1.



On page 7.

### WHAT IS IT LIKE TO WORK?

#### Full weekends off

The share of weeks with both Saturday and Sunday off. Today's link gives 4 in 20, one weekend in 5; this sheet gives 6 in 24, one weekend in 4.

**What it does not tell you.** A weekend off beside a 42-hour week is still a weekend off; the hours by week are on the grid, page 4.



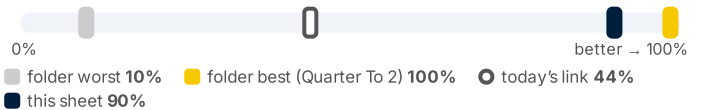
On page 5.

### WHAT IS IT LIKE TO WORK?

#### Working weeks on one turn

The share of working weeks where a person keeps the same clock time all week. A week that mixes an early and a late is what people mean by "all over the place".

**What it does not tell you.** One turn all week can still be a bad week if the turn is long; the week-to-week move and the duty lengths are on pages 5 and 8.



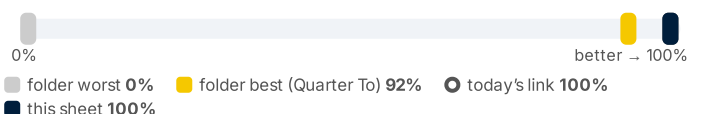
On page 5.

### IS IT FAMILIAR?

#### Shift times people work today

The share of this sheet's clock times that somebody on the 20-line link already works. It is the folder's proxy for how much there is to learn and to accept.

**What it does not tell you.** Familiar is not the same as popular — a time everyone works and nobody likes still counts — and a new time in one week is not the same as one in ten. The duty table on page 5 lists the new ones.



On page 5.

## BEFORE THE FIGURES

# The decision this sheet supports

What is being decided, which criteria are hard and which are the room's, how much each figure can bear, and what no page here can tell you. Read this before the tiles.

- 1 What this is
- 2 How to read the numbers
- 3 The decision it supports (this page)
- 4 The 24-line grid, with the words explained
- 5 A week compared with today
- 6 People on duty, hour by hour
- 7 Hard limits and the December 2026 figures
- 8 The ORR fatigue factors
- 9 Where it came from
- 10 The grid, ready to paste into the app

## The question

Should the link for the December 2026 timetable keep today's shift times, accepting one more cover cycle and nothing new to learn, or move to a table built from the December 2026 timetable rules and the timetable?

**Same Turns** is one of the three cases in the folder for keeping today's times; the others are *Quarter To 2 (Q2-24-W21)* and *Quarter To (QT-24-Q34)*. It is read against **By the Book (BB-24-D7)**, the one of the three rules-first sheets whose weekday cover follows the trains most closely (weekday fit 30.8) — not against every design in the folder.

## The criteria, in three tiers

### HARD — A DESIGN MEETS THESE OR CANNOT BE RUN

- ✓ Twelve hours between duties — met, tightest 13h 30m
- ✓ Chiltern's 13-day limit — longest run 6 days
- ✓ The 35-hour contract, exactly — 35h 00m

### SOFT — THE ROOM WEIGHS THESE AGAINST EACH OTHER

- December 2026 headcounts met: **4 of 4** (today 0 of 4)
- Fatigue factors present: **1** (today 5)
- Demand fit weekday · Sat · Sun: **46.3** · 32.7 · 80.6 (today 51.1 · 45.7 · 65.9; lower is closer)
- Times worked today: **15 of 15** · full weekends off: **6** in 24 (today 4 in 20)

### PREFERENCE — STAFF'S TO STATE, NOT THE TOOL'S

- Early against late: earlies 7h 25m–8h 30m, lates 7h 30m–9h 10m
- No new clock time to learn
- The longest duty, 9h 10m, and who holds it
- Cover weeks at lines 1, 7, 13, 19

## How much to trust each figure

| FIGURE ON THESE PAGES                                       | WHERE IT COMES FROM                                                                          | WEIGHT                                      |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------|
| 0 rests under 12h · 35h 00m contract · 14/10 at the weekend | Computed from the cells; re-computed by the workspace on import                              | <b>Firm</b>                                 |
| Longest run 6 days, inside 13                               | Computed; the 13-day limit's policy citation is still outstanding                            | <b>Firm figure</b> , unconfirmed threshold  |
| Four at the open, three to close, 14 and 10 at the weekend  | Owner-relayed December 2026 headcounts with no document behind them (class C)                | <b>Provisional</b>                          |
| 1 fatigue factor present                                    | The ORR's 2021 list, which it states are not limits; 4 factors still "definition to confirm" | <b>Advisory only</b> — never pass or fail   |
| Demand fit 46.3 · 32.7 · 80.6                               | Measured timetable, one share-based measure on one December 2026 timetable curve             | <b>Indicative</b>                           |
| 15 of 15 times are today's                                  | Computed against the live 20-line link                                                       | <b>Firm</b> , but a proxy for acceptability |
| Search score 8268                                           | The annealer's own objective; page 9 explains it                                             | <b>Do not weigh</b>                         |

## What no page here can tell you

- Whether the people who would work them accept the week shapes as written — familiarity is measured as "times worked today", which is a proxy for acceptability, not acceptability.
- Whether the roster clerk splits a cover week day-on-day-off. It is the one thing that moves the 48-hour-break figure (9 as a block, 10 split), and the link does not decide it.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 1, 7, 13, 19); nothing here models absence.
- Whether the December 2026 timetable demand curve holds on Wembley event days — the Saturday weighting approximates them, and the curve is a measured timetable, not a footfall count.
- Why Sunday's fit (80.6) is worse than today's (65.9) and whether anyone minds: the December 2026 headcount of ten is met by adding people, not by reshaping the day.

## Before it is frozen

Lines 6, 12, 16 hold the longest duty (9h 10m); the people who would work those weeks read them first and say in one line each what they would change. The roster office answers the cover-week question. Someone confirms the source of the 13-day limit. Only then does the factor count on page 8 mean anything, and only then is this sheet more than an aid to a conversation.

THE ROTATION

# The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

| LINE    | SUN               | MON               | TUE               | WED               | THU               | FRI               | SAT               | MON-SAT<br>hours | ALL WEEK<br>hours | DAYS WORKED<br>Mon-Sat | TURNS<br>Mon-Fri |
|---------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|-------------------|------------------------|------------------|
| 1       | SP                | SP                | SP                | SP                | SP                | SP                | SP                | 35h 00m          | 35h 00m           | —                      | —                |
| 2       | RD                | RD                | RD                | 08:00<br>16:30    | 08:00<br>16:30    | 08:00<br>16:30    | 06:20<br>14:50    | 34h 00m          | 34h 00m           | 4                      | <b>1</b>         |
| 3       | 07:15<br>15:45    | 08:00<br>16:30    | RD                | RD                | 08:00<br>16:30    | 08:00<br>16:30    | 06:20<br>14:00    | 33h 10m          | 41h 40m           | 4                      | <b>1</b>         |
| 4       | 07:15<br>15:45    | RD                | RD                | 06:20<br>13:45    | 06:20<br>13:45    | 06:20<br>13:45    | RD                | 22h 15m          | 30h 45m           | 3                      | <b>1</b>         |
| 5       | RD                | 13:30<br>22:00    | 13:30<br>22:00    | 13:30<br>22:00    | 13:30<br>22:00    | RD                | 14:30<br>22:00    | 41h 30m          | 41h 30m           | 5                      | <b>1</b>         |
| 6       | 14:30<br>23:25    | 15:15<br>23:55    | 15:15<br>23:55    | RD                | RD                | 15:15<br>23:55    | 14:45<br>23:55    | 35h 10m          | 44h 05m           | 4                      | <b>1</b>         |
| 7       | SP                | SP                | SP                | SP                | SP                | SP                | SP                | 35h 00m          | 35h 00m           | —                      | —                |
| 8       | RD                | RD                | 06:20<br>14:20    | 06:20<br>14:20    | 06:20<br>14:20    | 06:20<br>14:20    | RD                | 32h 00m          | 32h 00m           | 4                      | <b>1</b>         |
| 9       | RD                | 06:20<br>13:45    | 06:20<br>13:45    | 06:20<br>13:45    | 06:20<br>13:45    | RD                | RD                | 29h 40m          | 29h 40m           | 4                      | <b>1</b>         |
| 10      | RD                | 06:20<br>14:20    | 08:00<br>16:30    | 08:00<br>16:30    | 06:20<br>14:20    | 06:20<br>14:20    | RD                | 41h 00m          | 41h 00m           | 5                      | 2                |
| 11      | RD                | 15:15<br>23:55    | 15:15<br>23:55    | 15:15<br>23:55    | 15:15<br>23:55    | 15:15<br>23:55    | RD                | 43h 20m          | 43h 20m           | 5                      | <b>1</b>         |
| 12      | RD                | 15:15<br>23:55    | 15:15<br>23:55    | 15:15<br>23:55    | 15:15<br>23:55    | RD                | 14:45<br>23:55    | 43h 50m          | 43h 50m           | 5                      | <b>1</b>         |
| 13      | SP                | SP                | SP                | SP                | SP                | SP                | SP                | 35h 00m          | 35h 00m           | —                      | —                |
| 14      | 11:00<br>19:30    | RD                | RD                | 14:00<br>22:30    | 14:00<br>22:30    | 14:00<br>22:30    | 12:00<br>20:00    | 33h 30m          | 42h 00m           | 4                      | <b>1</b>         |
| 15      | 11:00<br>19:30    | RD                | RD                | 15:15<br>23:55    | 15:15<br>23:55    | 15:15<br>23:55    | 14:30<br>22:00    | 33h 30m          | 42h 00m           | 4                      | <b>1</b>         |
| 16      | 14:30<br>23:25    | 14:00<br>22:30    | RD                | RD                | 14:00<br>22:30    | 14:00<br>22:30    | 14:45<br>23:55    | 34h 40m          | 43h 35m           | 4                      | <b>1</b>         |
| 17      | RD                | 08:00<br>16:30    | 08:00<br>16:30    | RD                | RD                | 08:00<br>16:30    | 06:20<br>14:50    | 34h 00m          | 34h 00m           | 4                      | <b>1</b>         |
| 18      | 07:15<br>15:45    | 06:20<br>13:45    | 06:20<br>13:45    | RD                | RD                | 06:20<br>13:45    | 08:00<br>16:30    | 30h 45m          | 39h 15m           | 4                      | <b>1</b>         |
| 19      | SP                | SP                | SP                | SP                | SP                | SP                | SP                | 35h 00m          | 35h 00m           | —                      | —                |
| 20      | RD                | RD                | 13:30<br>22:00    | 13:30<br>22:00    | 13:30<br>22:00    | 13:30<br>22:00    | RD                | 34h 00m          | 34h 00m           | 4                      | <b>1</b>         |
| 21      | RD                | 08:00<br>16:30    | 08:00<br>16:30    | 08:00<br>16:30    | 08:00<br>16:30    | RD                | 06:20<br>14:50    | 42h 30m          | 42h 30m           | 5                      | <b>1</b>         |
| 22      | 07:15<br>15:45    | 06:20<br>14:20    | 06:20<br>14:20    | 06:20<br>14:20    | RD                | RD                | 08:00<br>16:30    | 32h 30m          | 41h 00m           | 4                      | <b>1</b>         |
| 23      | 14:30<br>23:25    | 13:30<br>22:00    | 14:00<br>22:30    | RD                | RD                | 13:30<br>22:00    | 14:00<br>22:30    | 34h 00m          | 42h 55m           | 4                      | 2                |
| 24      | 13:00<br>21:00    | 14:00<br>22:30    | 14:00<br>22:30    | 14:00<br>22:30    | RD                | RD                | 14:45<br>23:55    | 34h 40m          | 42h 40m           | 4                      | <b>1</b>         |
| Cover   | <b>10</b><br>SP 4 | <b>14</b><br>SP 4 | <b>14</b><br>SP 4 | <b>14</b><br>SP 4 | <b>14</b><br>SP 4 | <b>14</b><br>SP 4 | <b>14</b><br>SP 4 | 35h 00m          | 38h 34m           | 4.20                   | 18<br>one turn   |
| Minutes | 5,145             | 6,980             | 6,980             | 6,980             | 6,980             | 6,980             | 7,100             | 42,000           | 47,145            |                        |                  |

Early turns
Late turns
Cover (spare) week — four duties of seven, any turn
Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: distinct clock times Monday to Friday, **bold** where the week is one turn (one clock time Mon-Fri and one early-or-late family across every worked day — the page 1 figure) · Minutes: duty minutes per day, and Mon-Sat is the contract (20 × 35h = 42,000)

## The words on this page

**Link** — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

**Turn** — a shift time, such as 06:20-13:45. An **early** starts before 09:00; a **late** starts after.

**Cover (spare) week** — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

**The contract** — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

**Line** — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

**Rest day (RD)** — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

**Sunday** — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon-Sat and all week.

**One-turn week** — the same clock time all week (shown bold in the Turns column); the easiest kind of week to live around.

# Same feel, more people

What the search was told to keep, and what the December 2026 headcounts made it change

## Kept — the shape of a week

| MEASURE                                                  | TODAY (20)          | PROPOSED (24)             |
|----------------------------------------------------------|---------------------|---------------------------|
| Working weeks that are one turn (one clock time Mon–Fri) | 7 of 16             | <b>18 of 20</b>           |
| Weeks mixing early and late turns                        | 7                   | <b>0</b>                  |
| Rest-day breaks that are two days or more                | 13 of 17            | <b>20 of 24</b>           |
| Single rest days between duties (not a 48h break)        | 4                   | <b>4</b>                  |
| Days worked in a week (lines × days)                     | 1×3, 3×4, 11×5, 1×6 | <b>6×4, 14×5</b>          |
| Cover (spare) weeks (whole weeks)                        | lines 1, 7, 12, 17  | <b>lines 1, 7, 13, 19</b> |
| Distinct shift times (all already on today's roster)     | 18                  | <b>15</b>                 |
| Mon–Sat hours, average week (the contract)               | 35h 00m             | <b>35h 00m</b>            |
| Sunday duties (not contracted; half the working lines)   | 8 of 16 lines       | <b>10 of 20 lines</b>     |

## Changed — the December 2026 headcount

| PEOPLE ON DUTY                                            | TODAY     | PROPOSED         | WHY                                                             |
|-----------------------------------------------------------|-----------|------------------|-----------------------------------------------------------------|
| Monday to Friday                                          | 11–12     | <b>14</b>        | the contract: 20 working lines × 35h has to be worked somewhere |
| Saturday                                                  | 10        | <b>14</b>        | owner's figure for December 2026, leaning late for events       |
| Sunday                                                    | 8         | <b>10</b>        | owner's figure for December 2026                                |
| Opening at 06:20 (07:15 Sunday) wk · Sat · Sun            | 4 · 4 · 3 | <b>4 · 4 · 4</b> | four to open, every day — met                                   |
| Through to the close (23:55; 23:25 Sunday) wk · Sat · Sun | 2 · 2 · 3 | <b>3 · 4 · 3</b> | three to close, four on a Saturday — met                        |
| Still on duty at 22:00 wk · Sat · Sun                     | 4 · 2 · 3 | <b>5 · 5 · 3</b> | five from 22:00, Monday to Saturday — met                       |

## The duty table, in today's times

| TURN                                  | WK           | SAT          | SUN          | WK           | SAT          | SUN          |
|---------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| <del>06:20–13:35 7h 15m · Early</del> | <del>4</del> |              |              | <del>4</del> |              |              |
| 06:20–13:45 7h 25m · Early            | 1            |              |              | 2 +1         |              |              |
| 06:20–14:00 7h 40m · Early            |              | 1            |              |              | 1            |              |
| 06:20–14:20 8h 00m · Early            | 2            |              |              | 2            |              |              |
| 06:20–14:50 8h 30m · Early            |              | 3            |              |              | 3            |              |
| 07:15–15:45 8h 30m · Early            |              |              | 3            |              |              | 4 +1         |
| 08:00–16:30 8h 30m · Early            | 2            |              |              | 3 +1         | 2 +2         |              |
| <del>08:30–16:30 8h 00m · Early</del> |              |              | <del>4</del> |              |              | <del>4</del> |
| 11:00–19:30 8h 30m · Late             | 1            |              |              | -1           |              | 2 +2         |
| 12:00–20:00 8h 00m · Late             |              | 1            |              |              | 1            |              |
| 13:00–21:00 8h 00m · Late             |              |              | 1            |              |              | 1            |
| <del>13:30–21:00 7h 30m · Late</del>  |              | <del>4</del> |              |              | <del>4</del> |              |
| 13:30–22:00 8h 30m · Late             | 1            |              |              | 2 +1         |              |              |
| 14:00–22:30 8h 30m · Late             | 2            |              |              | 2            | 1 +1         |              |
| 14:30–22:00 7h 30m · Late             |              | 2            |              |              | 2            |              |
| 14:30–23:25 8h 55m · Late             |              |              | 3            |              |              | 3            |
| 14:45–23:55 9h 10m · Late             |              | 2            |              |              | 4 +2         |              |
| 15:15–23:55 8h 40m · Late             | 2            |              |              | 3 +1         |              |              |

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

Why the extra duties land where they do: a weekday needs 6,980 minutes and a Saturday 7,100 for 5 weekdays + 1 Saturday to total exactly  $20 \times 35h = 42,000$  — an equality, so the table is one of the few combinations of today's turn lengths that reaches it to the minute. The 06:20–13:35 turn is folded into 06:20–13:45 (ten minutes, one fewer time to remember); Sunday gains an 11:00–19:30 middle turn from the weekday vocabulary.

# People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

How to read it: each blue cell is the number of people on duty in that hour (darker blue = more people). The orange row is the timetable — how much of the day's trains that hour carries (darker orange = busier). Read the columns top to bottom: cover should be thickest where the orange is darkest. The **fit** figure at the right end of each row scores that match for the whole day.

## Monday to Friday window 06:20–23:55 · one row per distinct weekday

|                       | 05 | 06 | 07  | 08  | 09  | 10 | 11 | 12 | 13 | 14 | 15 | 16  | 17  | 18  | 19  | 20 | 21 | 22 | 23 | FIT  |
|-----------------------|----|----|-----|-----|-----|----|----|----|----|----|----|-----|-----|-----|-----|----|----|----|----|------|
| TODAY MON · WED       |    | 4  | 4   | 6   | 6   | 6  | 7  | 7  | 7  | 7  | 7  | 7   | 5   | 5   | 5   | 4  | 4  | 4  | 2  | 55.8 |
| TODAY TUE · THU · FRI |    | 4  | 4   | 6   | 6   | 6  | 7  | 7  | 8  | 8  | 8  | 8   | 6   | 6   | 6   | 5  | 5  | 4  | 2  | 50   |
| PROPOSED              |    | 4  | 4   | 7   | 7   | 7  | 7  | 7  | 9  | 9  | 10 | 10  | 7   | 7   | 7   | 7  | 7  | 5  | 3  | 46.3 |
| DEC 2026 TRAFFIC      | 5  | 69 | 107 | 127 | 122 | 95 | 85 | 75 | 84 | 79 | 87 | 105 | 140 | 138 | 107 | 90 | 73 | 75 | 66 |      |

## Saturday window 06:20–23:55

|                  | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | FIT  |
|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|
| TODAY            |    | 4  | 4  | 4  | 4  | 4  | 4  | 5  | 6  | 9  | 6  | 6  | 6  | 6  | 6  | 5  | 4  | 2  | 2  | 45.7 |
| PROPOSED         |    | 4  | 4  | 6  | 6  | 6  | 6  | 7  | 7  | 13 | 10 | 10 | 8  | 8  | 8  | 7  | 7  | 5  | 4  | 32.7 |
| DEC 2026 TRAFFIC | 6  | 24 | 58 | 75 | 81 | 92 | 88 | 77 | 64 | 68 | 72 | 84 | 82 | 67 | 64 | 79 | 78 | 62 | 41 |      |

## Sunday window 07:15–23:25

|                  | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | FIT  |
|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|
| TODAY            |    |    | 3  | 4  | 4  | 4  | 4  | 4  | 5  | 8  | 8  | 5  | 4  | 4  | 4  | 4  | 3  | 3  | 3  | 65.9 |
| PROPOSED         |    |    | 4  | 4  | 4  | 4  | 6  | 6  | 7  | 10 | 10 | 6  | 6  | 6  | 6  | 4  | 3  | 3  | 3  | 80.6 |
| DEC 2026 TRAFFIC |    |    | 15 | 45 | 65 | 71 | 74 | 71 | 61 | 55 | 64 | 69 | 78 | 73 | 75 | 52 | 71 | 65 | 50 |      |

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. **Fit** is one number for how closely the people on duty follow the trains through the day: 0 would be a perfect match, lower is better, and today's link scores what it scores. (For the record: the squared gap between each hour's share of the day's traffic and its share of the cover, inside the day's window, measured on duty minutes — so an hour a duty only touches counts in proportion; the cells above count heads.) It is the measure the searched tables were chosen on, and the fit of the average weekday is the *Wk fit* on page 9. Sunday: 5 December 2026 timetable movements fall after the 23:25 finish (the last at 23:54, three of them arrivals) — the standing question on whether Sunday's window moves; the window is stored per design, so the proposal can be rebuilt to either answer.

**Reading it.** The proposal raises the whole day rather than reshaping it, which is what the December 2026 timetable brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The one hour the search was allowed to favour is Saturday's 17:00–22:00, for events. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

# The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

**Hard limits** — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **6** days (today: 9)  
Longest possible run is 6 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows. Worked as a BLOCK of four rather than split day-on-day-off, the answer is the same — **6** days — so this row does not depend on how a cover week is placed.  
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)  
Tightest anywhere: **13h 30m — line 14 Fri 14:00-22:30 into Sat 12:00-20:00** (today's tightest: 12h 30m). The generator refuses a design it cannot repair to this; the search here never produced one.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon–Sat over the 24 lines, cover weeks counted as contracted weeks  
700h of duty a week across 20 working lines. Sundays (85.75h) sit on top as RDW, as they do today. Individual weeks range 22h 15m–43h 50m; only the average is the contract.

**December 2026 timetable design figures** — the staffing shape agreed for the new timetable

| RULE                                           | PROPOSAL                                                                                                                                                             |                                                                                                                        |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Four on at the open, every day                 | ✓ 4 weekday · 4 Saturday · 4 Sunday                                                                                                                                  |                                                                                                                        |
| Three through to the close; four on a Saturday | ✓ 3 weekday · 4 Saturday · 3 Sunday                                                                                                                                  |                                                                                                                        |
| Five still on duty at 22:00                    | ✓ 5 weekday · 5 Saturday                                                                                                                                             | a 22:00 finish is not "on at 22:00"                                                                                    |
| Fourteen on a Saturday, ten on a Sunday        | ✓ 14 · 10                                                                                                                                                            |                                                                                                                        |
| Four cover weeks, evenly spread                | ✓ lines 1, 7, 13, 19 — gaps 6, 6, 6, 6                                                                                                                               | whole weeks, never scattered days                                                                                      |
| About 4.2 days a week worked, Mon–Sat          | ✓ 4.20 over the 20 working lines                                                                                                                                     |                                                                                                                        |
| No :05 or :10 times except the open and close  | ✓ none                                                                                                                                                               | today's times already obey this                                                                                        |
| Late turns slightly shorter than most earlies  | × weekday not met (lates 8h30–8h40, earlies 7h25–8h30) · Saturday not met (lates 7h30–9h10, earlies 7h40–8h30) · Sunday not met (lates 8h00–8h55, earlies 8h30–8h30) | incompatible with keeping today's times: meeting it needs 9h+ earlies or 15 duties a day — flagged for decision, below |

## To settle before it is frozen

**Late-turn length.** The one December 2026 preference this proposal does not meet is the owner's own — lates slightly shorter than most earlies. Today's link has it the other way round (a 15:15–23:55 late is 8h40, a 06:20–14:20 early 8h00) and the brief was to keep today's times; meeting both is arithmetically impossible at 14 duties a day, because the day's minutes are fixed by the contract. The choice is between today's clock times and shorter lates paid for by longer earlies (the workspace default does this, with 9h25 earlies). **Sunday's finish** — five December 2026 timetable movements fall after 23:25; the proposal inherits today's window deliberately rather than deciding it.

## ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021).  $\Delta$  present — the pattern is in this design and worth a look, not a breach ·  $\checkmark$  clear — it is not ·  $\bullet$  standing — true of the station itself, not of any design · — does not apply here. This is the list the link is assessed against.

The Office of Rail and Road lists 25 roster patterns that tend to tire people — long runs of earlys, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question for the room, never a pass or a fail, and a design showing nothing is not thereby approved.

| CODE | FACTOR                                                                                               | TODAY'S LINK                                        | PROPOSED                                             |
|------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------|
| FF1  | Night shift covering 00:00–05:00 <small>Time of day</small>                                          | --                                                  | --                                                   |
| FF2  | Early shift starting 05:00–07:00 <small>Time of day</small>                                          | $\bullet$ 24                                        | $\bullet$ 24                                         |
| FF3  | Very early shift starting before 05:00 <small>Time of day</small>                                    | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF4  | Very early shift (before 05:00) over 8h <small>Duty length</small>                                   | --                                                  | --                                                   |
| FF5  | Day shift over 12h <small>Duty length</small>                                                        | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF7  | Early shift over 10h <small>Duty length</small>                                                      | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF8b | Fewer than 2 days rest after a block of early starts <small>Recovery time</small>                    | $\Delta$ 3                                          | $\checkmark$ 0                                       |
| FF11 | More than 13 consecutive shifts without a 48h break <small>Recovery time</small>                     | $\checkmark$ 12<br>$\checkmark$ 9 as rostered       | $\checkmark$ 10<br>$\checkmark$ 9 as rostered        |
| FF10 | More than 4 consecutive 12h day shifts <small>Cumulative</small>                                     | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF15 | More than 4 consecutive early shifts in a rotating pattern <small>Cumulative</small>                 | $\Delta$ 6                                          | $\checkmark$ 4                                       |
| MRSF | More than 12 consecutive day shifts <small>Cumulative</small>                                        | $\checkmark$ 9                                      | $\checkmark$ 6                                       |
| MRSF | More than 55 hours worked in any 7-day period <small>Cumulative</small>                              | $\Delta$ 60.8                                       | $\checkmark$ 51                                      |
| MRSF | More than 7 consecutive 8h shifts (definition to confirm) <small>Cumulative</small>                  | $\Delta$ 9                                          | $\checkmark$ 6                                       |
| MRSF | More than 6 consecutive night or early shifts in a permanent pattern <small>Cumulative</small>       | --                                                  | --                                                   |
| FF17 | Backward rotating pattern (definition to confirm) <small>Circadian</small>                           | $\checkmark$ 10 backward / 12 forward               | $\checkmark$ 13 backward / 13 forward                |
| FF18 | Rotating pattern of about a week (definition to confirm) <small>Circadian</small>                    | $\bullet$ 20-line rotation · typically 4h 0m a week | $\bullet$ 24-line rotation · typically 2h 33m a week |
| FF19 | Successive start times varying by more than 2 hours (definition to confirm) <small>Circadian</small> | $\Delta$ 6                                          | $\Delta$ 1                                           |
| FF6  | Night shift over 10h <small>Night shifts</small>                                                     | --                                                  | --                                                   |
| FF8  | Less than 2 days rest after consecutive nights <small>Night shifts</small>                           | --                                                  | --                                                   |
| FF9  | Less than 14h rest in any 24h for night shifts <small>Night shifts</small>                           | --                                                  | --                                                   |
| FF12 | Only one day rest after night shifts <small>Night shifts</small>                                     | --                                                  | --                                                   |
| FF14 | More than 4 consecutive nights in a rotating pattern <small>Night shifts</small>                     | --                                                  | --                                                   |
| FF16 | More than 3 consecutive night shifts over 8h <small>Night shifts</small>                             | --                                                  | --                                                   |
| FF20 | First night shift <small>Night shifts</small>                                                        | --                                                  | --                                                   |
| FF13 | Less than 12h rest in any 24h <small>Intervals between duties</small>                                | $\checkmark$ 0                                      | $\checkmark$ 0                                       |

Present: 1 (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a BLOCK, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

# How it was chosen

Reproducible: the same table, seed and objective give the same grid on any machine

**Lineage.** Family ST — the existing 20-line link kept in its own shift times and week shapes. Compare with “Five Shift Times” (23 Aug 2026, family FST) and the workspace's own December 2026 duty-table default.

**In plain terms.** A computer took each day's duties as fixed and tried over a million different ways of arranging them across the 24 weeks, scoring every arrangement on the rules on pages 7 and 8, and kept the best. The “seed” is the number that makes that run repeat exactly on any machine, so anyone can rebuild this grid and get the same cells.

## How it was chosen

**1 • The table.** Every possible December 2026 duty table was enumerated from today's shift times — every count of each turn, on each day class — and kept only if it opened with four, closed with three (four on Saturday), kept five on at 22:00, and totalled exactly 42,000 Mon–Sat minutes. 81 tables survive. Two weekday shapes were carried forward — table A with three 11:00 middles, table B with a third 08:00 and today's 13:30 turn — and every finished design was scored against the December 2026 timetable demand curve; B, used here, follows a weekday's curve more evenly than today's roster does (51.1 today, 46.3 here, lower is more even), where A follows it less well (63.7) — the three 11:00 middles put nine people on the quietest part of the afternoon.

**2 • The grid.** Simulated annealing over the 24-line grid: 60,000 candidate arrangements per restart, four restarts from different random starting points per run, three seeded runs per table — over a million candidates in all — then a steepest-descent polish of every

same-day swap and every pair of lines. Every move keeps each day's duties exactly as the table says — a swap between two lines on one day, or two whole lines changing places — so the coverage, the contract and the headcounts are true by construction and only the SHAPE is searched. Cover weeks stay at lines 1, 7, 13 and 19 throughout.

**3 • The judge.** Every candidate was scored by the workspace's own modules — runDesignChecks, assessFatigue, assessHardLimits, scoreOrder — with a rest under 12h or a run past six days costing more than everything else combined, fatigue factors next, and then the feel terms: a week on more than one turn, a single rest day between duties, fewer than four or more than five days, and the week-to-week step. Weekends off and long weekends were rewarded. Last, the app's line-order optimiser was offered the result and kept only where it improved that score.

## Alternatives measured

| DESIGN                                                        | RUN | FF PRESENT | WKENDS | ONE-TURN | STEP   | WK FIT | SCORE   |
|---------------------------------------------------------------|-----|------------|--------|----------|--------|--------|---------|
| ST-24-B7 · d15e1b74 — <b>Same Turns (this proposal)</b>       | 6   | 1          | 6      | 18/20    | 2h 33m | 46.3   | 8268    |
| ST-24-A21 · 235ecdd7 — runner-up                              | 6   | 1          | 6      | 18/20    | 1h 47m | 63.7   | 6742    |
| ST-24-A13 · 22966914                                          | 6   | 1          | 5      | 17/20    | 2h 31m | 63.7   | 10806   |
| ST-24-B21 · 8a3198a8                                          | 6   | 2          | 6      | 18/20    | 3h 8m  | 46.3   | 11528   |
| ST-24-A7 · 320073c5                                           | 6   | 2          | 6      | 19/20    | 3h 31m | 63.7   | 11566   |
| ST-24-B13 · adaa379c                                          | 6   | 2          | 5      | 18/20    | 2h 26m | 46.3   | 11826   |
| Workspace default · 29ec9810 (Dec 2026 table, generated)      | 7   | 5          | 5      | 0/20     | 4h 16m | 30.8   | 2192436 |
| BB-24-D7 · 0f14abce — <b>By the Book</b> (a sibling proposal) | 6   | 0          | 6      | 12/20    | 2h 52m | 30.8   | —       |
| Today's 20-line link (for scale)                              | 9   | 5          | 4      | 7/16     | 4h 0m  | 51.1   | —       |

**How the winner was picked:** rules first (rest, run, fatigue factors present, weekends off), then *Wk fit* — how evenly the weekday cover follows the December 2026 timetable traffic curve (lower is better; today's link scores what it scores). The top two tied on every rule; the fit decided it (46.3 against 63.7), and the runner-up is named so it can be asked for. Score is the search's own feel objective (lower is better) and is not a verdict. The workspace default is the app's own December 2026 duty table, generated and reordered with every switch on — a good design by every panel, and not one that resembles today's.

LOAD IT

## The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Links → Import, then paste the whole block below. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied beside this PDF as Same-Turns-ST-24-B7-import.txt (tab-separated, pastes directly) and Same-Turns-ST-24-B7.json (the app's own format).

|    |             |             |             |             |             |             |             |
|----|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| 1  | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 2  | RD          | RD          | RD          | 08:00-16:30 | 08:00-16:30 | 08:00-16:30 | 06:20-14:50 |
| 3  | 07:15-15:45 | 08:00-16:30 | RD          | RD          | 08:00-16:30 | 08:00-16:30 | 06:20-14:00 |
| 4  | 07:15-15:45 | RD          | RD          | 06:20-13:45 | 06:20-13:45 | 06:20-13:45 | RD          |
| 5  | RD          | 13:30-22:00 | 13:30-22:00 | 13:30-22:00 | 13:30-22:00 | RD          | 14:30-22:00 |
| 6  | 14:30-23:25 | 15:15-23:55 | 15:15-23:55 | RD          | RD          | 15:15-23:55 | 14:45-23:55 |
| 7  | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 8  | RD          | RD          | 06:20-14:20 | 06:20-14:20 | 06:20-14:20 | 06:20-14:20 | RD          |
| 9  | RD          | 06:20-13:45 | 06:20-13:45 | 06:20-13:45 | 06:20-13:45 | RD          | RD          |
| 10 | RD          | 06:20-14:20 | 08:00-16:30 | 08:00-16:30 | 06:20-14:20 | 06:20-14:20 | RD          |
| 11 | RD          | 15:15-23:55 | 15:15-23:55 | 15:15-23:55 | 15:15-23:55 | 15:15-23:55 | RD          |
| 12 | RD          | 15:15-23:55 | 15:15-23:55 | 15:15-23:55 | 15:15-23:55 | RD          | 14:45-23:55 |
| 13 | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 14 | 11:00-19:30 | RD          | RD          | 14:00-22:30 | 14:00-22:30 | 14:00-22:30 | 12:00-20:00 |
| 15 | 11:00-19:30 | RD          | RD          | 15:15-23:55 | 15:15-23:55 | 15:15-23:55 | 14:30-22:00 |
| 16 | 14:30-23:25 | 14:00-22:30 | RD          | RD          | 14:00-22:30 | 14:00-22:30 | 14:45-23:55 |
| 17 | RD          | 08:00-16:30 | 08:00-16:30 | RD          | RD          | 08:00-16:30 | 06:20-14:50 |
| 18 | 07:15-15:45 | 06:20-13:45 | 06:20-13:45 | RD          | RD          | 06:20-13:45 | 08:00-16:30 |
| 19 | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 20 | RD          | RD          | 13:30-22:00 | 13:30-22:00 | 13:30-22:00 | 13:30-22:00 | RD          |
| 21 | RD          | 08:00-16:30 | 08:00-16:30 | 08:00-16:30 | 08:00-16:30 | RD          | 06:20-14:50 |
| 22 | 07:15-15:45 | 06:20-14:20 | 06:20-14:20 | 06:20-14:20 | RD          | RD          | 08:00-16:30 |
| 23 | 14:30-23:25 | 13:30-22:00 | 14:00-22:30 | RD          | RD          | 13:30-22:00 | 14:00-22:30 |
| 24 | 13:00-21:00 | 14:00-22:30 | 14:00-22:30 | 14:00-22:30 | RD          | RD          | 14:45-23:55 |

Prepared 8 September 2026 · re-rendered 25 September 2026. The code **d15e1b74** in every footer is the design's fingerprint — eight characters computed from these 168 cells and nothing else. If the workspace shows the same code after you paste, you have exactly this design; if one cell differs, the code changes. Every figure in this document was computed by the Marylebone Roster app's Links modules (v24.23) from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.